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Structural Performance Evaluation of Historical Bursa Green Mosque

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Abstract

The Ottoman Empire, which ruled Anatolia for more than 600 years, brought many religious, social and artistic structures to the region. In particular, some structures are equipped with unique features to display the power of the rulers and the splendor of the empire. However, most of these structures are damaged over time due to natural or human origin reasons. Since Turkey is located in an earthquake zone, these structures are under seismic risk. For these structures to be preserved correctly, first of all, their structural behavior should be examined and the parts that are at risk in terms of structural performance, if any, should be strengthened/repaired with appropriate techniques. In this study, Bursa Green Mosque, which exhibits the characteristic examples of early period Ottoman tile art with its technical and adornment features, was discussed and the structural performance of the mosque examined. Within the scope of the study, the structural finite element model of Bursa Yeşil Mosque, located in the Yeşil district of Bursa, was firstly prepared, and then the structural performance and seismic behavior of the structure were evaluated by using finite element analyses. In the finite element analyses static, modal and response spectrum analyses were conducted to determine structural performance.

Keywords: Bursa Green Mosque, Structural Performance, Seismic Behaviour, Finite Element Method, Static Analysis, Modal Analysis, Response Spectrum Analysis

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Introduction

Bursa, one of the most important cities of Turkey with its historical and cultural diversity, has hosted many civilizations throughout history. The city, which even served as the capital city in the early periods of the Ottoman Empire, has been able to carry the traces of many civilizations it hosted in the past to the present day. The Ottoman Empire, which ruled Anatolia for more than 600 years, brought many religious, social and artistic structures to the region. In particular, some structures are equipped with unique features in order to display the power of the rulers and the splendor of the empire. One of these structures is Bursa Green Mosque. This structure is located in the green district of Bursa province, on an elevated position overlooking the city. The mosque is actually a part of a social complex consisting of a tomb, mosque, madrasah and almshouse (Fig. 1). It was built in 1419 by Sultan Celebi Mehmet, son of Sultan Yıldırım Beyazıt. The entrance to the mosque is provided by the crown gate in the north. According to studies by (Pay, 1996: 181, Şevki 1933: 29); as cited in (Terzi Doğan, 2010: 15) from the inscription on the upper part of the crown gate it is understood that the architect of the mosque was Hacı İvaz Pasha (Fig. 2-3). It is likely that mosque got its name from the predominantly green tile adornments once found on its minarets. Evliya Çelebi, who visited Bursa in the 17th century, stated that the minaret of the mosque was covered with green tiles and that is why it was called the Green Mosque (Kurşun & Kahraman 2011: 13); as cited in (Budak, 2018: 71-72).



Fig. 1. Green Mosque and Green district of Bursa

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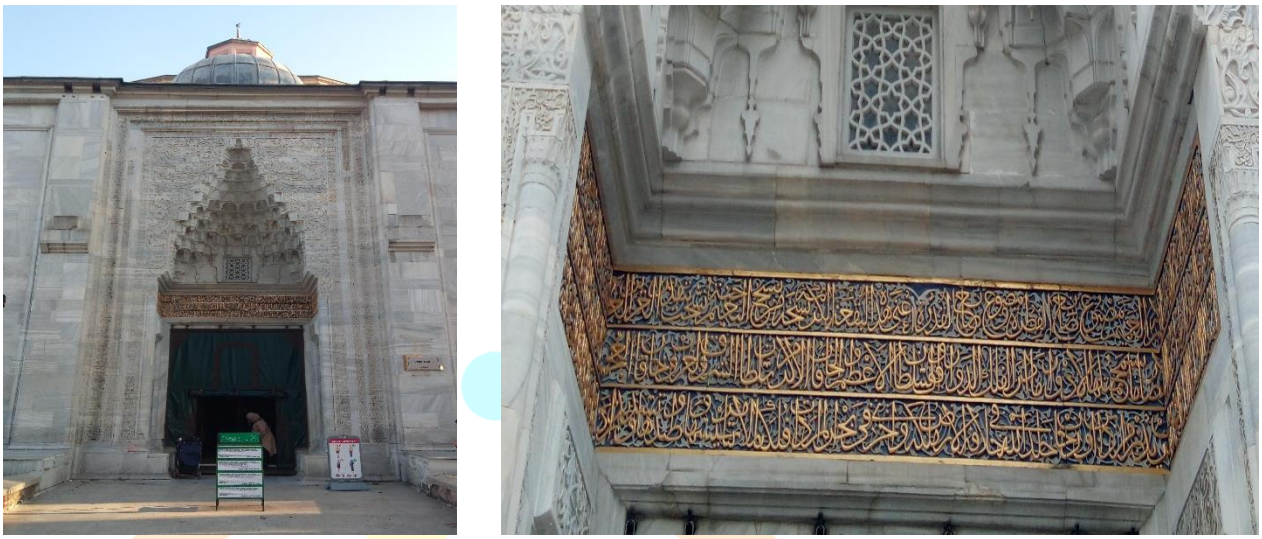


Fig. 2-3. The crown gate and the inscription on the crown gate

The plan of the mosque is an example of the winged mosques (Reverse T) type. The adjacent parts which are located on the east and west part of the prayer hall represent the wings (Fig. 4-5). The survey drawings showing the different facades of the mosque are given in Fig. 6 and Fig. 7.

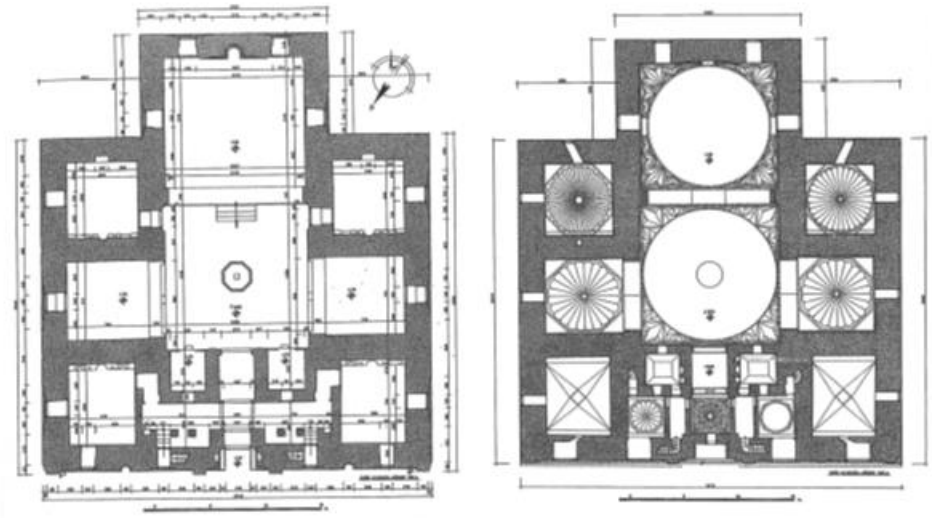


Fig. 4-5. The ground and first floor plan of the mosque (Ayverdi and Yüksel, 1976; as cited in Uslu, 2012: 19-20)

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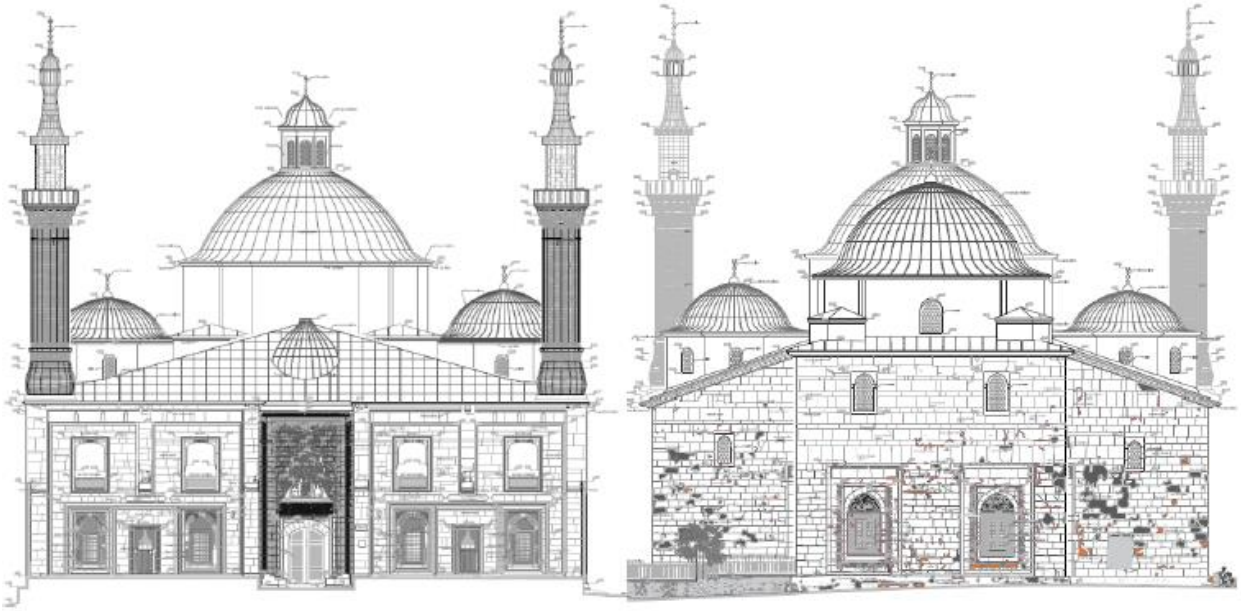


Fig. 6. Northern and southern facades of Bursa Green Mosque
(Bursa Vakıflar Bölge Müdürlüğü); as cited in (Mutlu, 2015: 50)

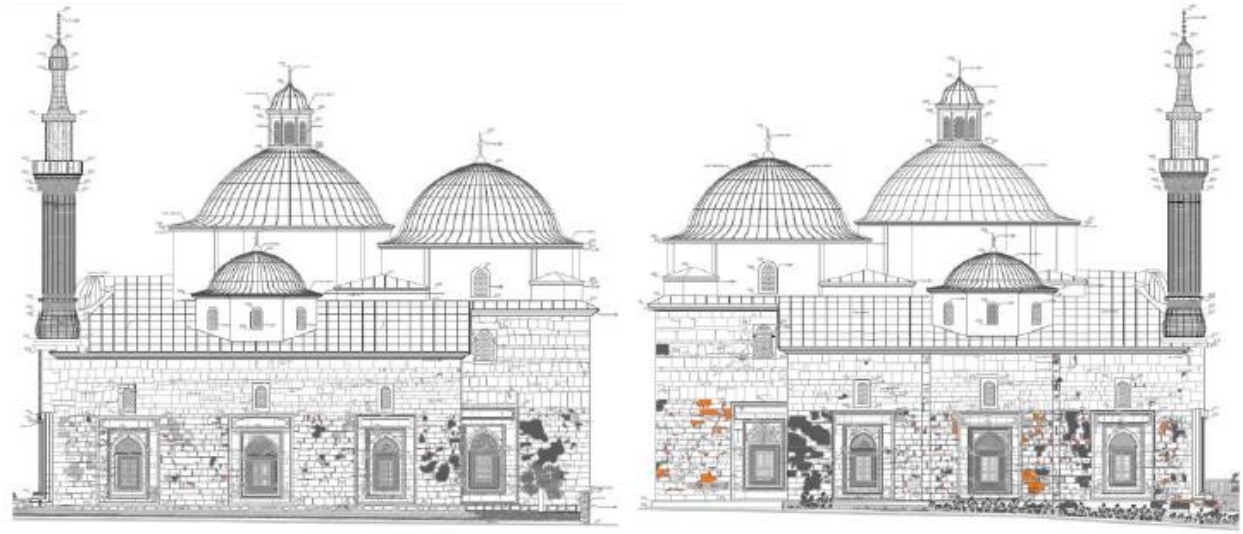


Fig. 7. Western and eastern facades of Bursa Green Mosque
(Bursa Vakıflar Bölge Müdürlüğü); as cited in (Mutlu, 2015: 50)

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If the sources up to the great Bursa earthquake of 1855 are examined; It is understood that the mosque has a single minaret (Budak, 2018: 72-73). Today, the mosque has two minarets in the north-east and north-west corners (Fig.8-9). The available data reveal that the eastern minaret was added in 1863 during the restoration made by the French architect Leon Parvillée (Budak, 2018: 73). It is known that the minarets of the mosque, which have been repaired many times until today, were demolished in the late 19th century and both were rebuilt during the reign of the governor Reşid Mümtaz Pasha (Pay 2010: 55); as cited in (Budak, 2018: 71).



Fig. 8-9. North-east and north-west minarets of the mosque

The Current Situation and Structural Features of Bursa Green Mosque

The structure generally consists of two large domed halls on the mihrab axis, two iwans to the right and left of them with smaller domes relatively, and rectangular rooms with lead roofs at the corners (Fig 4-5). The mosque is entered through the crown gate, which is located in the north and has magnificent carved ornamental details (Fig. 2). The crown gate opens to the entrance hall. On the east and west sides of the entrance hall there are two corridors with a width of approximately 2.3m. There are two reused Byzantine spolias in each of these corridors, but it is not known when these spolias were added to the structure (Fig. 10). It is known that these columns were added to the structure later and even Evliya Çelebi also used a definite statement about the structure by saying "There are no columns in it" (Kurşun, Kahraman 2013: 13); as cited in (Budak, 2018: 74).

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Fig. 10. Details from Byzantine Spolias

The stairs in these corridors lead to the upper floor. Here, there are rooms called sultan mahfil and decorated with uniquely beautiful tiles (Fig. 11). After the entrance hall on the ground floor, a passage covered with a barrel vault is passed to reach the domed area with an octagonal fountain in the middle (Fig. 12) The dome above this area has a diameter of 13 m and a height of 25 m (Aslanapa 1986: 38); as cited in (Terzi Doğan, 2010: 21). In the middle of the dome there is a lantern. This dome is connected to the mihrab dome with a Bursa type arch (Fig. 13). There is no lighting lantern in mihrab dome.



Fig. 11. Sultan mahfil (upper one) and muezzin mahfils (lower ones)

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Fig. 12. Octagonal fountain under the main dome



Fig. 13. Bursa type arch connecting the two main domes

The iwans in the north east and north west are connected to the structure by arches. The tops of these iwans are covered by domes with 24 slices, which are relatively smaller than the main domes. The arch and dome of the iwan in the west are shown in Fig. 14. The mihrab of the mosque is covered with magnificent tiles and carved ornaments (Fig. 15)

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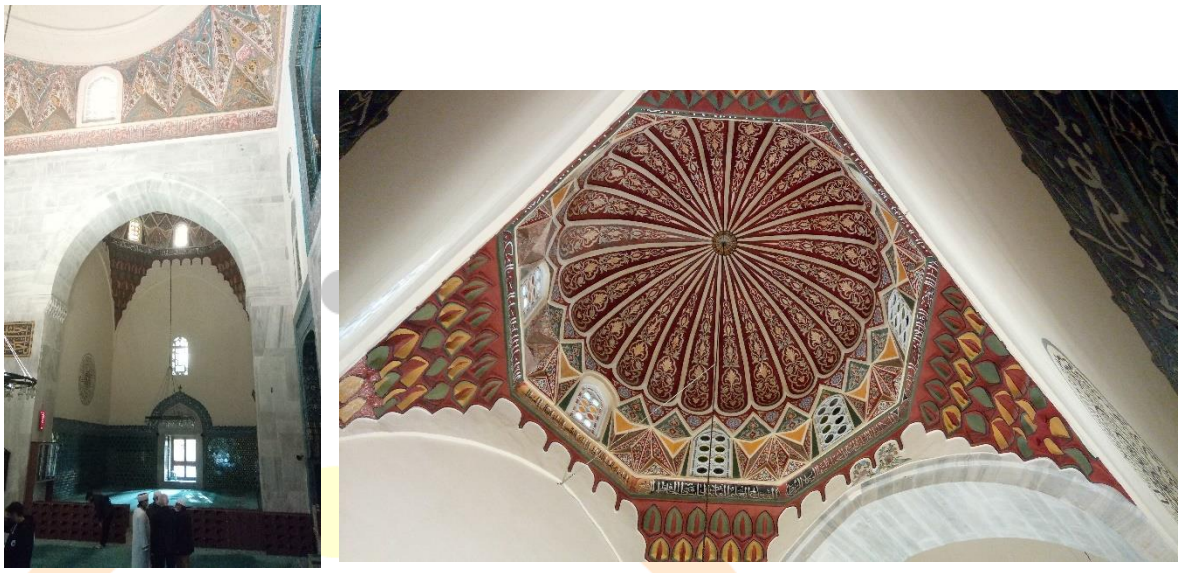


Fig. 14. The arch and dome of the iwan in the west

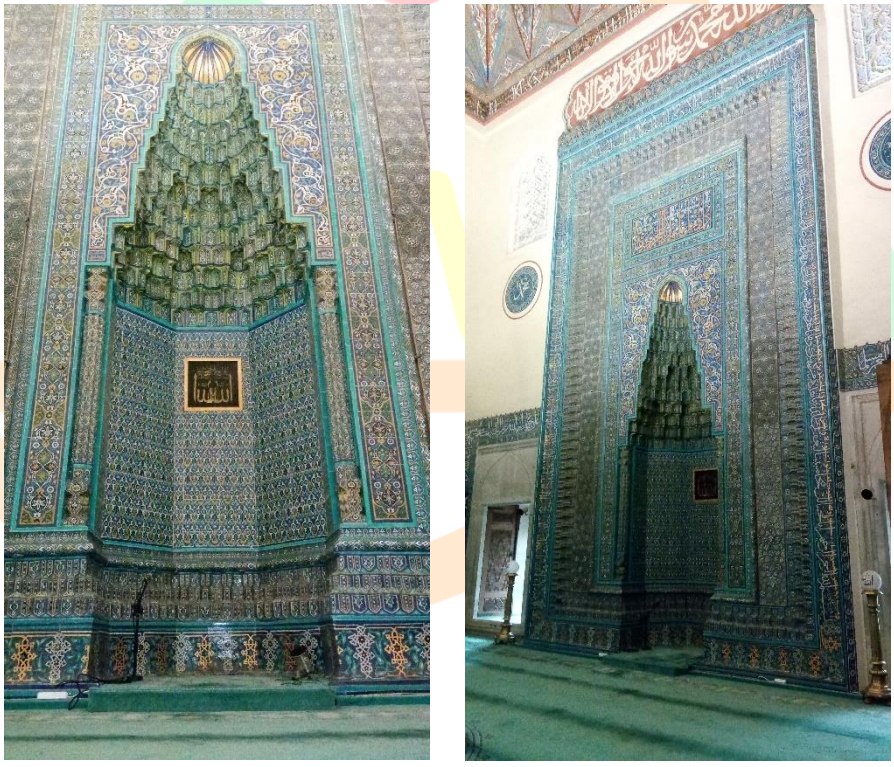


Fig. 15. The mihrab of the mosque

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Smooth cut stone was used on the walls of the Green Mosque. To cover these walls marble was used intensely inside and outside the building. The northern facade of the building is completely covered with marble (Fig. 16).



Fig. 16. The northern facade of the mosque

METHODS

Within the scope of the study, to determine the structural performance of the historical Bursa Green Mosque, a 3D finite elements model of the building was prepared and structural analyzes were made using ANSYS Workbench program. The following assumptions and parameters have been taken into account in the finite element model prepared and the analyzes made:

1) While creating the models, the drawings of Bursa Regional Directorate of Foundations were used (Mutlu, 2015: 50)

2) The historical mosque is modeled in 3D using 474642 nodal points and 272678 solid elements.

3) The technical properties of the materials used in the model have been chosen from the values suggested in the literature, taking into account the studies on similar structures and academic studies. The modulus of elasticity is 10000 MPa for the stone forming the main walls of the building, and 13000 MPa for the brick used as the covering material; Poisson's ratio is 0.22 for stone and 0.16 for brick; Density is taken as 2200 kg/m3 for stone and 2200 kg/m3 for brick (Dabanlı, 2008; Korkmaz et al.; 2014); as cited in (Mutlu, 2015: 106).

4) In the dynamic analysis phase, response spectrum analysis was performed.

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5) In the finite element analysis, it is assumed that the materials behave linearly-elasticly.

6) The results obtained are shown with colored stress distributions and graphics in the ANSYS Workbench program.

Static Analysis

In the structural analysis phase, first of all, the behavior of the structure under its own weight was examined, for this purpose static analyzes were carried out. The finite element model with meshes is shown in Fig.17. The total vertical displacements obtained as a result of the analysis are shown in Fig.18. As can be seen from the figure, the maximum vertical displacement occurred at the top of the middle dome and its size was approximately 1,15 mm. The maximum principal stresses are concentrated in the middle of the spans and arches, its maximum value is approximately 0,47 MPa. (Fig. 19). It is seen that the minimum principal stresses are concentrated at the places where the arches are attached to the structure and at the points where the building rests on the ground, their maximum value is approximately 1,81 MPa.

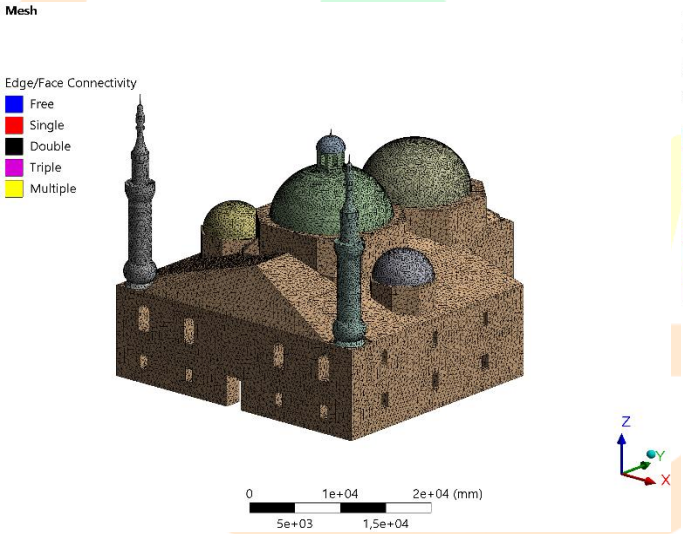


Fig. 17. Finite element model of the mosque

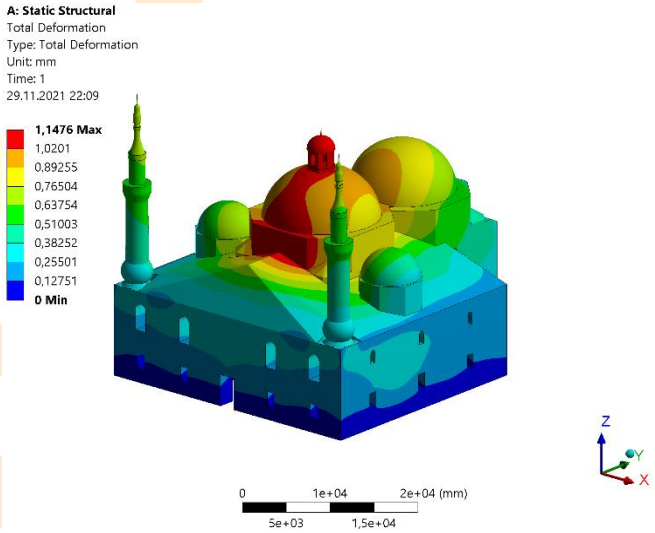


Fig. 18. Total vertical displacements

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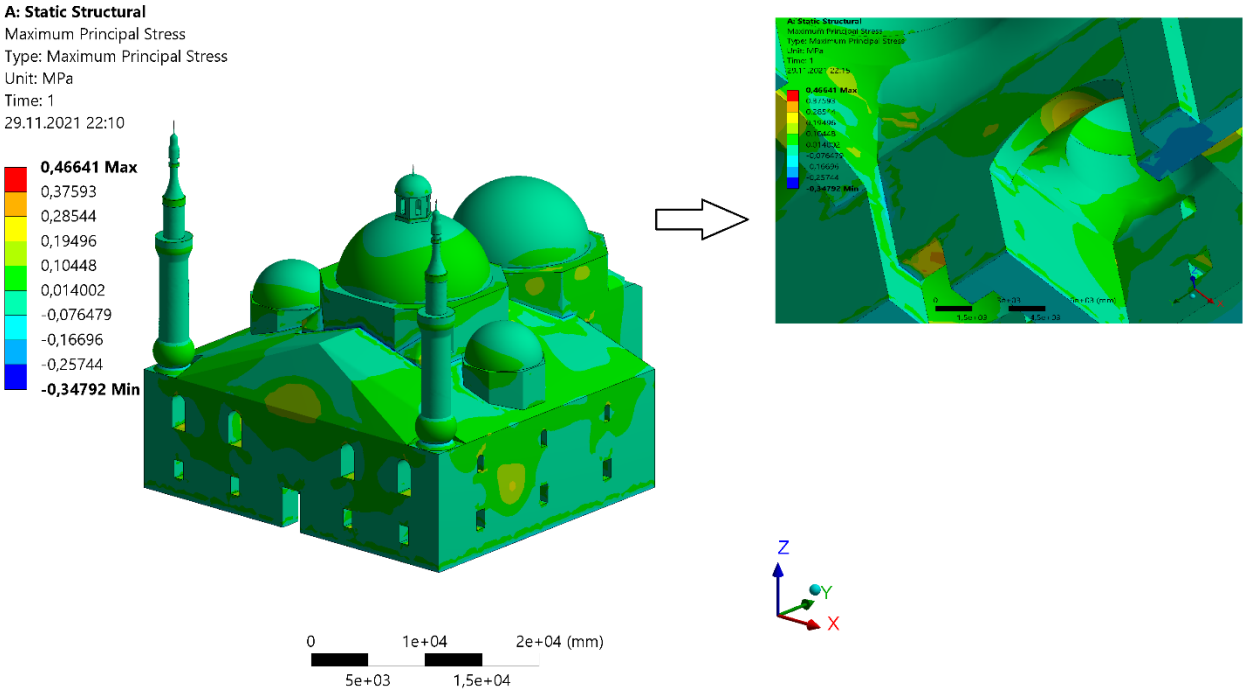


Fig. 19. Maksimum Principal Stresses

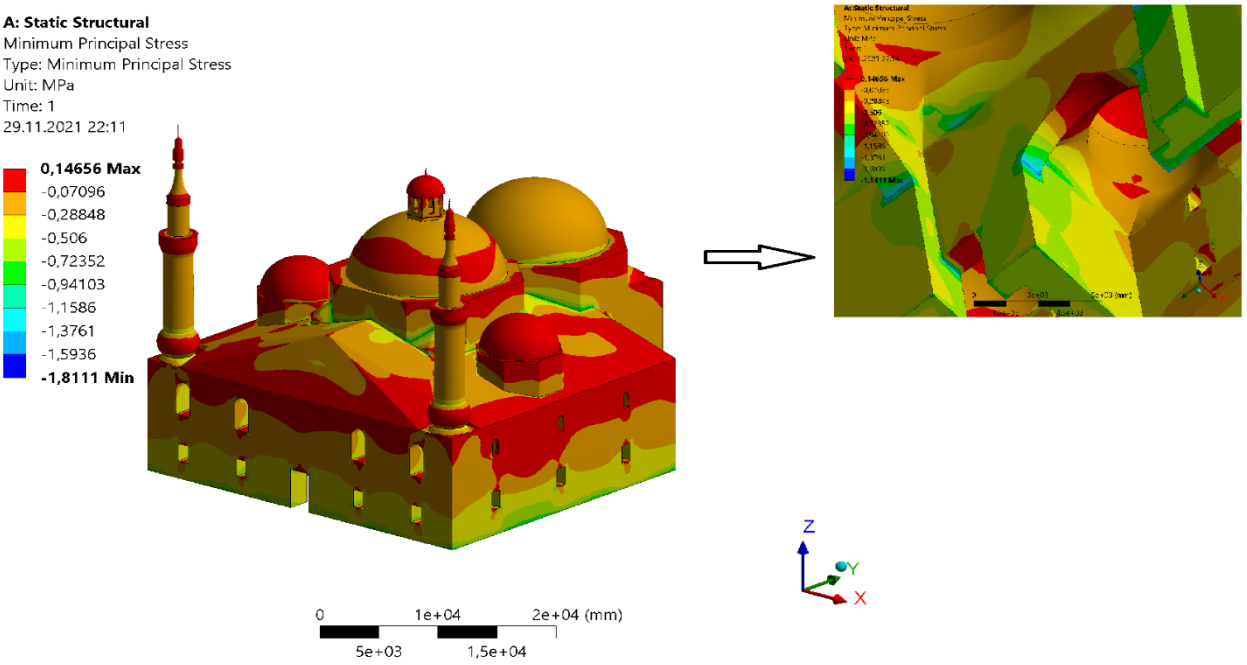


Fig. 20. Minimum Principal Stresses

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Modal Analysis

In the dynamic analysis of Bursa Green Mosque, modal analyzes were carried out to determine the free vibration forms of the structure. Thus, mode shapes and vibration periods that determine the dynamic character of the structure are obtained. It is expected that the largest displacements in the structure and therefore the dominant modes will occur in the minarets. This may affect the interpretation of the overall behavior of the structure. For this reason, two different modal analyzes were made for the building, with and without minaret. The mode shapes of the first six modes for the analysis without minaret are shown in Fig. 21. The frequency values and mass participation rates of the first six modes are also shown in Fig. 22. for the analysis without minaret (for X, Y and Z axes).

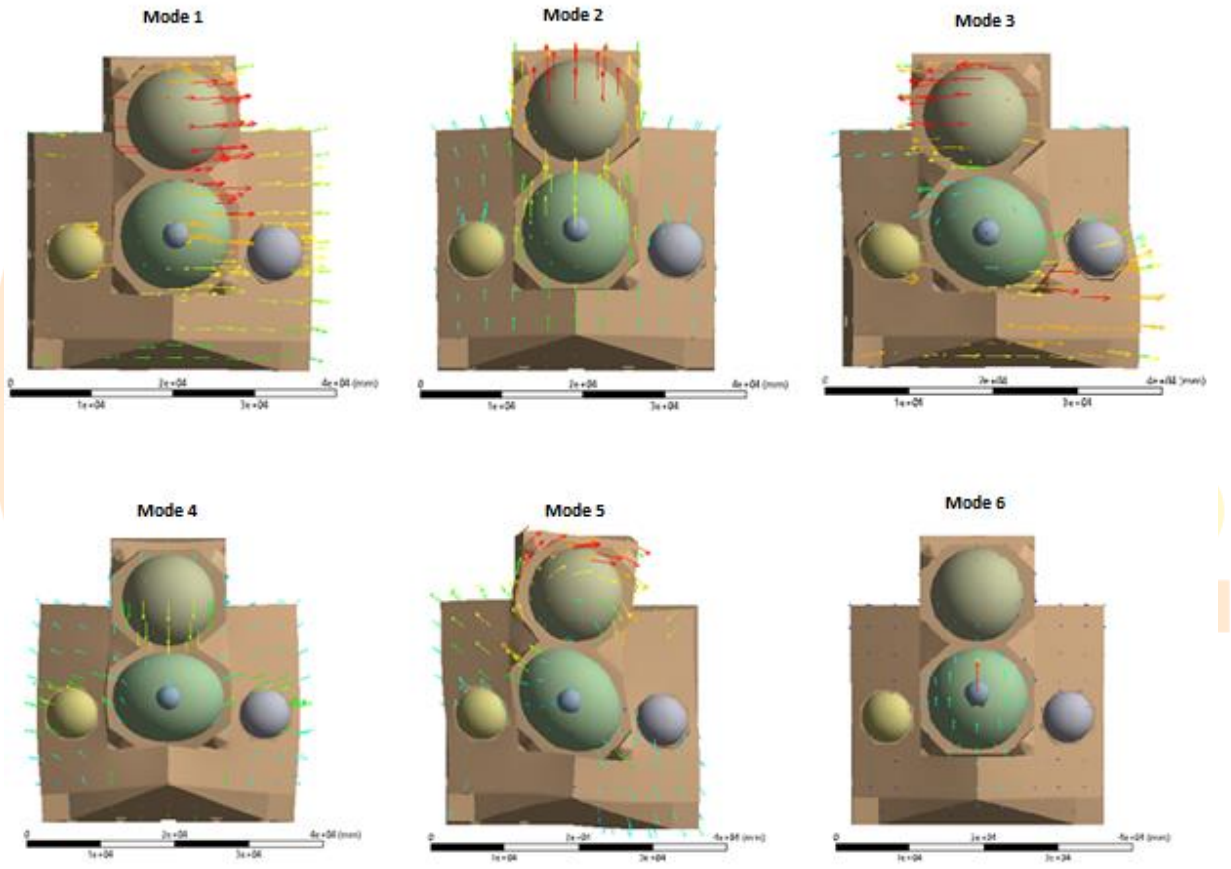


Fig. 21. First six mode shapes for structure (without minaret)

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***** PARTICIPATION FACTOR CALCULATION ***** X DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	9.61730	0.10398	109.61	1.000000	12013.9	0.979753	0.745001
2	9.82375	0.10179	-2.5647	0.023399	6.57754	0.980289	0.407884E-03
3	12.5065	0.79958E-01	15.303	0.139619	234.193	0.999388	0.145227E-01
4	13.6390	0.73319E-01	0.26491E-01	0.000242	0.701774E-03	0.999388	0.435181E-07
5	14.7764	0.67675E-01	2.7389	0.024988	7.50139	1.00000	0.465174E-03
6	17.4077	0.57446E-01	-0.44111E-01	0.000402	0.194581E-02	1.00000	0.120663E-06

***** PARTICIPATION FACTOR CALCULATION ***** Y DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	9.61730	0.10398	2.2943	0.022408	5.26366	0.475317E-03	0.326408E-03
2	9.82375	0.10179	102.39	1.000000	10483.1	0.947116	0.650073
3	12.5065	0.79958E-01	0.60919	0.005950	0.371117	0.947149	0.230135E-04
4	13.6390	0.73319E-01	23.610	0.230595	557.430	0.997486	0.345671E-01
5	14.7764	0.67675E-01	-0.10830E-01	0.000106	0.117279E-03	0.997486	0.727266E-08
6	17.4077	0.57446E-01	5.2762	0.051532	27.8385	1.00000	0.172631E-02

***** PARTICIPATION FACTOR CALCULATION ***** Z DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	9.61730	0.10398	0.68414E-01	0.003143	0.468045E-02	0.683754E-05	0.290242E-06
2	9.82375	0.10179	0.16777	0.007708	0.281456E-01	0.479547E-04	0.174535E-05
3	12.5065	0.79958E-01	0.61259E-02	0.000281	0.375270E-04	0.480095E-04	0.232711E-08
4	13.6390	0.73319E-01	-14.516	0.666904	210.716	0.307877	0.130668E-01
5	14.7764	0.67675E-01	0.33467E-01	0.001538	0.112004E-02	0.307878	0.694558E-07
6	17.4077	0.57446E-01	21.766	1.000000	473.773	1.00000	0.293794E-01

Fig. 22. The frequency values and mass participation rates of the first six modes (without minaret)

The mode shapes of the first six modes for the analysis with minarets are shown in Fig. 23. The frequency values and mass participation rates of the first six modes are also shown in Fig. 24. for the analysis with minarets (for X, Y and Z axes).

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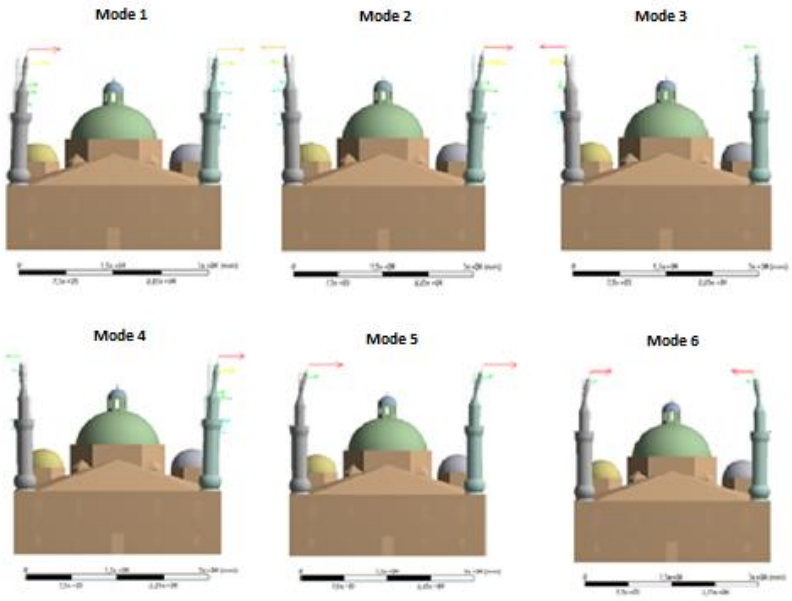


Fig. 23. First six mode shapes for structure (with minarets)

***** PARTICIPATION FACTOR CALCULATION ***** X DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	3.36930	0.29680	12.172	0.468399	148.165	0.165525	0.900579E-02
2	3.37521	0.29628	1.0656	0.041004	1.13547	0.166793	0.690166E-04
3	3.41639	0.29271	-7.9707	0.306717	63.5315	0.237768	0.386159E-02
4	3.41919	0.29247	2.6371	0.101478	6.95431	0.245537	0.422699E-03
5	8.54600	0.11701	-25.987	1.000000	675.326	0.999989	0.410479E-01
6	8.58428	0.11649	0.99443E-01	0.003827	0.988898E-02	1.00000	0.601075E-06

***** PARTICIPATION FACTOR CALCULATION ***** Y DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	3.36930	0.29680	0.82692	0.049579	0.683801	0.141443E-02	0.415630E-04
2	3.37521	0.29628	-9.8151	0.588475	96.3367	0.200685	0.585557E-02
3	3.41639	0.29271	3.2255	0.193385	10.4035	0.222205	0.632351E-03
4	3.41919	0.29247	9.8907	0.593009	97.8269	0.424558	0.594615E-02
5	8.54600	0.11701	-0.93019E-01	0.005577	0.865252E-02	0.424576	0.525920E-06
6	8.58428	0.11649	-16.679	1.000000	278.186	1.00000	0.169088E-01

***** PARTICIPATION FACTOR CALCULATION ***** Z DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	3.36930	0.29680	-0.11135E-01	0.093647	0.123983E-03	0.767815E-02	0.753596E-08
2	3.37521	0.29628	0.11890	1.000000	0.141376E-01	0.883209	0.859317E-06
3	3.41639	0.29271	0.10745E-01	0.090372	0.115464E-03	0.890359	0.701815E-08
4	3.41919	0.29247	0.37431E-01	0.314807	0.140108E-02	0.977127	0.851611E-07
5	8.54600	0.11701	-0.92120E-02	0.077476	0.848612E-04	0.982383	0.515806E-08
6	8.58428	0.11649	0.16866E-01	0.141852	0.284478E-03	1.00000	0.172912E-07

Fig. 24. The frequency values and mass participation rates of the first six modes (with minarets)

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Dynamic Analysis

In the dynamic analysis phase, response spectrum analysis was performed for Bursa Green Mosque. In the analysis, first of all, the necessary data were obtained from the web site of AFAD by using the coordinates of the location of the Bursa Green Mosque. The ground class for the location of the mosque was chosen as ZB (slightly weathered, moderately strong rocks). Earthquake ground motion level was chosen as DD-2 (PGA with 10% probality of exceedance in 50 years/475 year return period). As a result of the modal analysis, the dominant period of the structure was found on the x-axis (in the direction of the short side of the structure). Two different response spectrum analyzes were made for the building, with and without minaret. As a result of the analyses, the lateral displacements obtained for the cases with and without a minaret are given in Fig. 25 and Fig. 26, respectively. As can be seen from the figures, the maximum lateral displacement occurred at the top of the minarets for the model with minarets and its size was approximately 76,4 mm. However the maximum lateral displacement occurred at the top of the domes for the model without minaret and its size was approximately 3,16 mm.

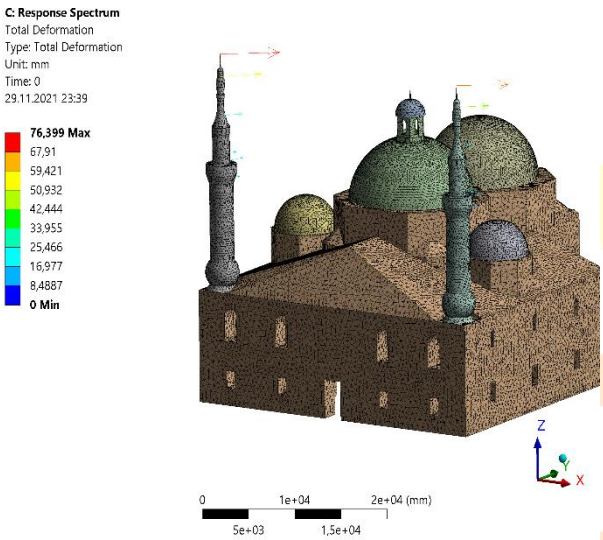


Fig. 25. Lateral displacements with minarets

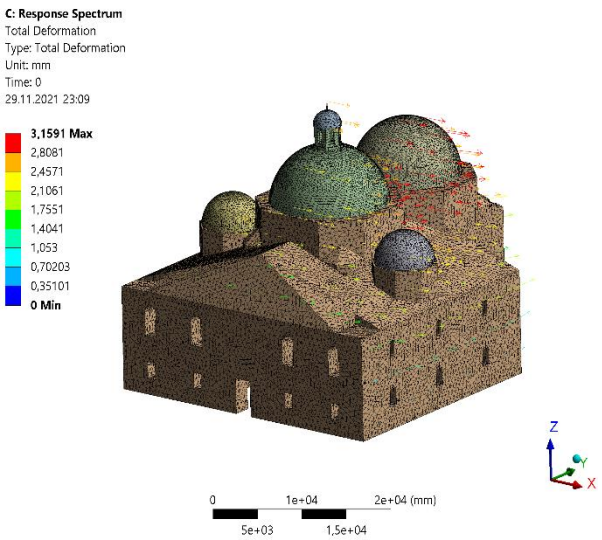


Fig. 26. Lateral displacements without minaret

Similarly, the normal stress distributions for the analyzes with and without minaret are given in Fig. 27 and Fig. 28, respectively.

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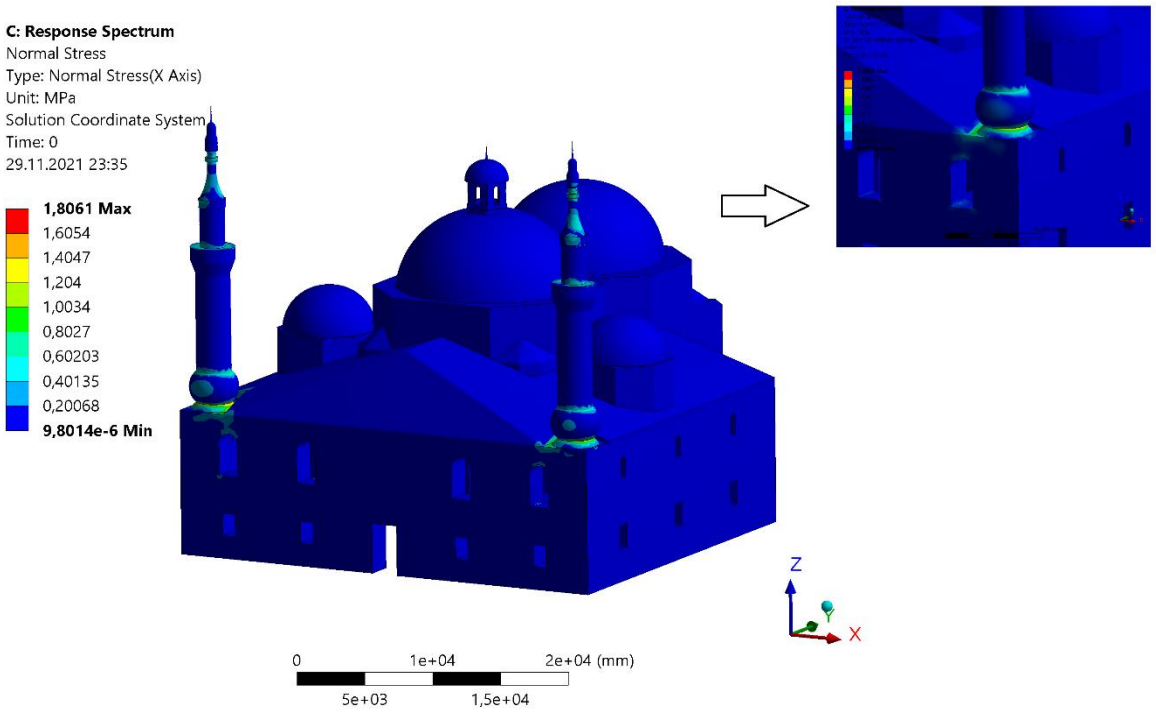


Fig.27 Normal stress distributions with minarets

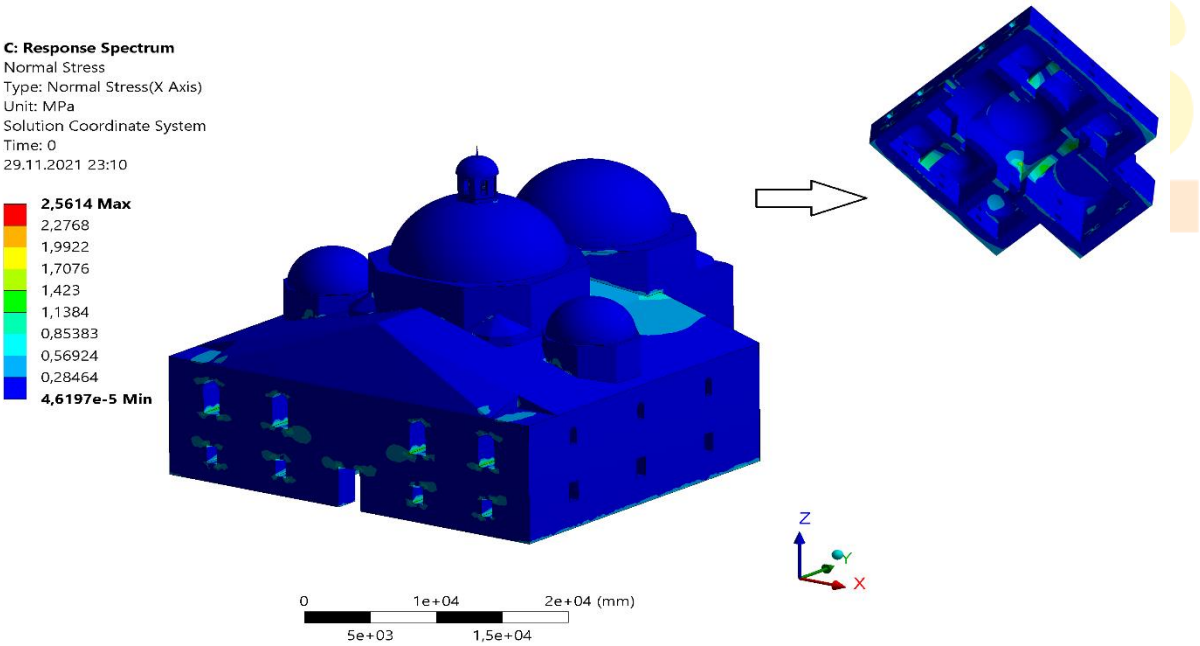


Fig.28. Normal stress distributions without minaret

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As expected for the model without minaret, the maximum normal stresses are concentrated in the middle of the spans and arches, its maximum value is approximately 2,56 MPa. For the model with minarets, the maximum normal stresses are concentrated at the points where the minarets attached to the structure (the points where the cross-section changes abruptly), with a maximum value of approximately 1,81 MPa.

After all these analyzes, as an acceptance criterion, it can be checked whether the calculated displacements are acceptable by comparing the limit situations in FEMA 356 (2000), FEMA 274 and ASCE 41 (2006) regulations with the current displacements.

Table 1. Acceptance criteria for unreinforced masonry in-plane walls and piers

		Primary members			Secondary members	
	Limiting behavioral mode	Immediate occupancy (IO) (%)	Life safety (LS) (%)	Collapse prevention (CP) (%)	Life safety (LS) (%)	Collapse prevention (CP) (%)
FEMA 356 (2000) and	Bed-joint sliding	0.1	0.3	0.4	0.6	0.8
FEMA 274	Rocking	0.1	$0.3h_{eff}/L$	$0.4h_{eff}/L$	$0.6h_{eff}/L$	$0.8h_{eff}/L$
ASCE 41 (2006)	Rocking	0.1	$0.3h_{eff}/L$	$0.4h_{eff}/L$	$0.6h_{eff}/L$	$0.8h_{eff}/L$

(Cakir et al., 2015: 197)

To find the drift ratio in the model without minaret; the maximum displacement (3.16mm) is divided by the building height (28414mm) to get 0.011%. As can be seen from the table above, in this case the structure is in the "Immediate Occupancy" zone.

To find the drift ratio in the model with minarets; the maximum displacement (76,4mm) is divided by the building height (32711mm) to get 0.023%. As can be seen from the table above, in this case the structure is now close to the "Life Safety" zone.

CONCLUSIONS AND RECOMMENDATIONS

The maximum stresses and displacements in the analyzes for Bursa Green Mosque occurred at the expected levels and expected places. This indicates the accuracy of the analysis performed. However, it has been observed that the minarets of the structure will displace more than the structure during the earthquake and that large stresses will occur in the places where the minarets attached to the structure, so these sections are risky parts in terms of earthquakes. In addition, in this study, the properties of the materials used in the building were taken from the data given for similar structures in the literature. For this reason, it is recommended to perform the necessary mechanical tests by taking various samples from the structure and to repeat the finite element analyzes with the data obtained.

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